

Shubham Gupta

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SUMMARY

Computer Science senior (Schreyer Honors College, GPA 3.83) with production AI systems experience across enterprise software, distributed infrastructure, and full-stack ML deployment. Paper accepted at **AIES 2026**. **AI Innovation Fellow at Deloitte** (Summer 2026). Open-source contributor to **pytorch/ignite** (29K★) and **sentence-transformers** (18.9K★).

EDUCATION

The Pennsylvania State University — Schreyer Honors College University Park, PA
B.E. in Computer Science, Minor in Artificial Intelligence Engineering; GPA: 3.83/4.00 Expected Dec. 2026

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Machine Learning, Computer Networks, Systems Programming
- Honors Thesis: Designed a three-way behavioral diagnostic decomposing LLM codebook-classification predictions into definition-following, label-name-matching, and surface-feature-matching, extending Halterman & Keith (2025); found RAG improves accuracy (+5.2% wF1) while increasing label-name compliance from 56% to 67.5% (k=1→k=10), and that compliance-preserving fine-tuning partially recovers definitional fidelity (0.285 vs. RAG's 0.155) though not to zero-shot levels (0.78). (Advisor: Prof. Qunhua Li)

PUBLICATIONS

Do Personalized Evaluation Functions Reflect Human Preferences? A Study of Weighted Proximity in Algorithmic Recourse. A. Shyamsundar Rekha, et al. (incl. S. Gupta). **Accepted at AIES 2026**, Malmö, Sweden, Oct. 2026.

EXPERIENCE

AI Innovation Fellow June 2026 – Aug. 2026
Deloitte Philadelphia, PA

- Selected for competitive AI Innovation program; deploying production AI systems using **LLMs**, **agentic architectures**, and **AWS infrastructure** for enterprise-scale applications.

Undergraduate Research Assistant Jan. 2026 – Present
Penn State Department of Statistics — Prof. Qunhua Li State College, PA

- Engineered end-to-end NLP pipeline for large-scale text annotation across CCC, BFRS, and Manifestos datasets on **ICDS Roar HPC** synthetic LoRA fine-tuning, RAG retrieval, and ablation experiments.
- Invited reviewer, **AIES-26 Program Committee** (invited by Prof. Aylin Caliskan, University of Washington); reviewed 3 papers on AI ethics and social impact.

IT Summer Associate June 2025 – Aug. 2025
University of Pittsburgh Medical Center (UPMC) Pittsburgh, PA

- Engineered **Nursing Matrix** application using **ASP.NET Core** and **C#**, integrating Epic EHR data across four hospital units serving 200+ staff; architected RESTful API layer with **Redis** caching, reducing query latency by **18%** under concurrent load.

TECHNICAL PROJECTS

CodeBase | *FastAPI, FAISS, sentence-transformers, AWS DynamoDB/S3, Groq LLaMA-3.3-70B, Next.js, Vercel* 2026

- Built AI-powered codebase intelligence tool semantic search over any GitHub repo using **sentence-transformers** embeddings + **FAISS** vector index; answers natural-language questions with file:line citations grounded in source code.
- Implemented multi-repo comparison enabling side-by-side architectural analysis across two codebases with per-repo citation attribution. Live at codebase-frontend.vercel.app.
- Production stack: **AWS DynamoDB** for chunk metadata, **S3** for FAISS indexes, **Groq LLaMA-3.3-70B** for generation, Next.js on **Vercel**.

Open Source — pytorch/ignite & sentence-transformers (29K★ / 18.9K★) | *Python, PyTorch, FastAPI, Sphinx* 2026

- Implemented **CharacterErrorRate** metric for `ignite.metrics.nlp` — character-level edit distance evaluation for ASR/OCR; 15 unit tests, full Sphinx docs. Merged PR [PR #3785](https://github.com/pytorch/ignite/pull/3785), shipped in v0.5.2.
- Authored production deployment guide for sentence-transformers covering async server (FastAPI/uvicorn) event-loop blocking and the correct fix pattern, following maintainer-guided root-cause correction. PR [PR #3856](https://github.com/sentence-transformers/sentence-transformers/pull/3856), under review.

SkinIQ | *React, TypeScript, FastAPI, Python, Docker, Supabase, PostgreSQL, Railway* 2026

- Architected full-stack web application with **React/TypeScript** frontend, **FastAPI** backend, and **PostgreSQL** with Row Level Security. Live at skin-iq.vercel.app.
- Integrated ML inference service with primary/fallback pipeline graceful degradation with <15s timeout and automatic retry logic.

LEADERSHIP

Founding Member / Mechanical Lead, Robot in 3 Days (Ri3D) Aug. 2023 – Present

- Led cross-functional team building functional competition robots within 72-hour hard deadlines; responsible for system architecture and component integration.

Mentor, FIRST Global Teams (Belize, Niger, Rwanda, Bolivia) June 2021 – Present

- Mentored 20+ students across four countries in software engineering, robotics, and systems design for international competition.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, JavaScript/TypeScript, SQL, Bash
AI/ML: PyTorch, LoRA fine-tuning, RAG pipelines, FAISS, sentence-transformers, Groq, LLM evaluation
Systems: Multithreading, Concurrency, POSIX, Sockets, Memory Management, Distributed Systems, REST APIs
Web/Infra: React, FastAPI, ASP.NET Core, PostgreSQL, Redis, Docker, AWS (DynamoDB, S3), Vercel, Git, Linux